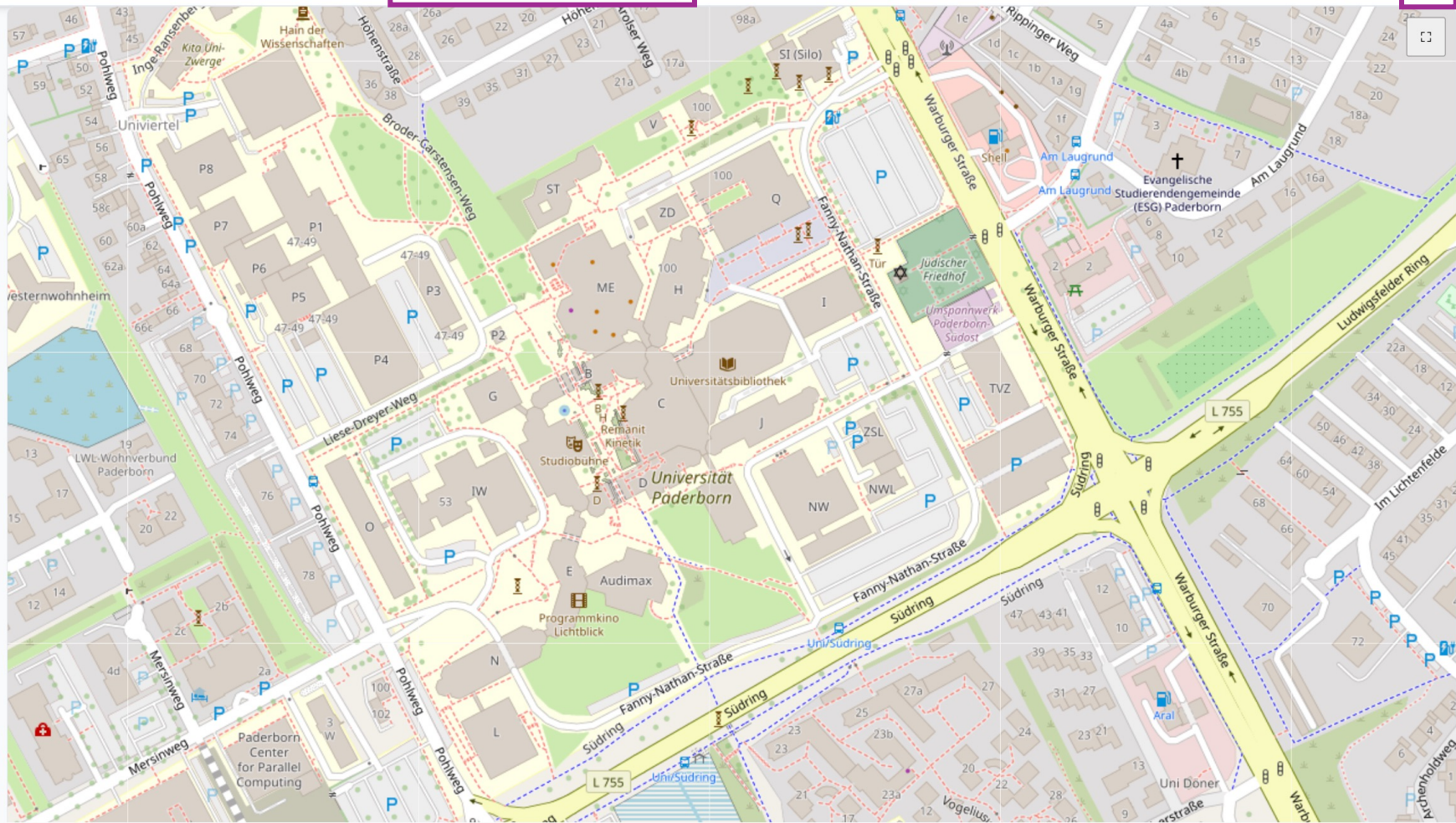


Lock map to start working



[+ Add Process](#)

 Save

LOAD

 Measure

 Export CSV

Placed successfully

OpenStreetMap

Process 1

Place

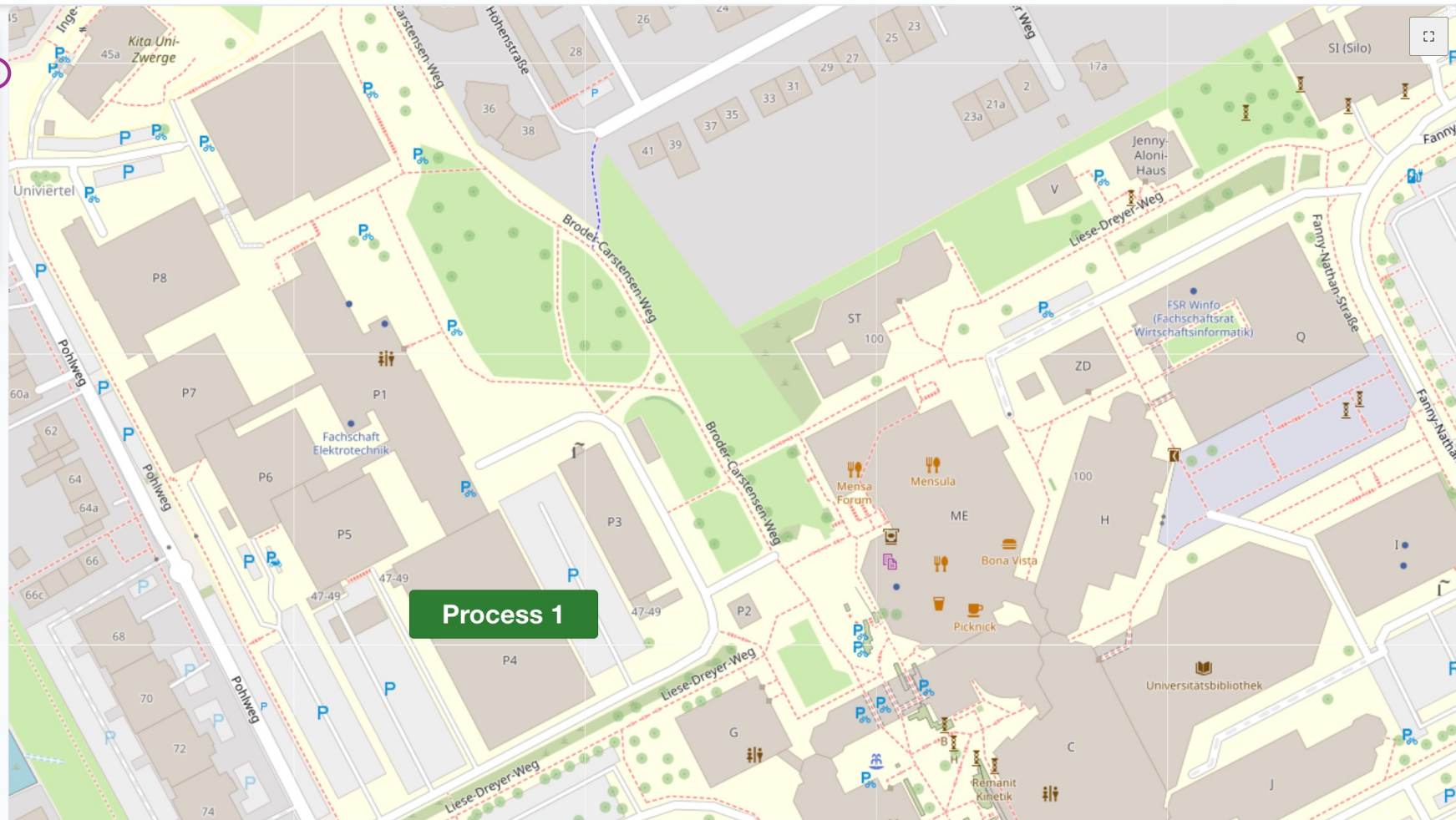
1 sub

×

Process 1

3

The screenshot displays the configuration interface for 'Process 1'. The 'Subprocesses (1)' section contains a 'Show on Map' button (labeled 1) and an 'Add' button (labeled 2). The 'Subprocess 1' section includes a 'Place' button and a close icon. The 'Next Processes (Click to link)' section features a 'Streams' tab (labeled 3) and a 'Notes' tab. The 'STREAMS (0)' section at the bottom has an 'Add Stream' button.



[Unlock Map](#)[Add Process](#)[Save](#)[LOAD](#)[Measure](#)[Export CSV](#)

Locked Map

[OpenStreet](#)

Process 1

Place 4 sub

Information

Subprocesses (4)

Hide Map

Add

Subprocess 1

1

Next Processes (Click to link)

Streams

Notes

STREAMS (2)

2

Add Stream

Steam Subprocess 1 s2

Steam

Tin Tout m cp CP Water Content In

Water Content Out Density Pressure

TIN °C M kg/s

103 20

Product Subprocess 1 s1

Product

Tin Tout m cp CP Water Content In

Water Content Out Density Pressure

TIN °C TOUT °C M kg/s

43 23 5

CP kJ/(kg·K)

3

Subprocess 2

Place

Subprocess 3

Place

Subprocess 4

Place



Potential Analysis

All streams

Scenario

Save

Save

LOAD

Report

Streams Selection

Subprocess 1

STEAM SUBPROCESS 1 S2

TIN:103°C | TOUT:43°C | CP:40.00 | Q:2400.00 KW

PRODUCT SUBPROCESS 1 S1

TIN:43°C | TOUT:23°C | CP:15.00 | Q:300.00 KW

Subprocess 2

PRODUCT SUBPROCESS 4 S3

TIN:128°C | TOUT:103°C | CP:15.00 | Q:375.00 KW

Subprocess 3

UTILITY SUBPROCESS 5 S4

TIN:88°C | TOUT:113°C | CP:30.00 | Q:750.00 KW

PRODUCT SUBPROCESS 1

TIN:68°C | TOUT:88°C | CP:50.00 | Q:1000.00 KW

Subprocess 4

SUBPROCESS 6 PRODUCT 6

TIN:8°C | TOUT:68°C | CP:20.00 | Q:1200.00 KW

Current energy supply

Supply 1

HEAT (KW)3000

▼ Cold streams selection

UTILITY SUBPROCESS 5 S4

PRODUCT SUBPROCESS 1

SUBPROCESS 6 PRODUCT 6

COOLING (KW)3000

▼ Hot streams selection

STEAM SUBPROCESS 1 S2

PRODUCT SUBPROCESS 1 S1

PRODUCT SUBPROCESS 4 S3

+ Add Supply

Comparison of Status Quo and Proposal

TYPE OF ENERGY DEMAND	CURRENT STATUS QUO (KW)	MIN. DEMANDS AFTER HEAT RECOVERY (KW)	SAVINGS (KW)	SAVINGS (%)	HP COVERAGE (KW) – VHTHP (HFC/HFO)	HEAT PUMP COVERAGE (%)
Heating Demand	3000.0	375.0	2625.0	87.5%	375.0	100.0%
Cooling Demand	3000.0	500.0	2500.0	83.3%	257.0	51.4%



Comparison of Status Quo and Proposal

TYPE OF ENERGY DEMAND	CURRENT STATUS QUO (KW)	MIN. DEMANDS AFTER HEAT RECOVERY (KW)	SAVINGS (KW)	SAVINGS (%)	HP COVERAGE (KW) – VHTHP (HFC/HFO)	HEAT PUMP COVERAGE (%)
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Cooling Demand	3000.0	500.0	2500.0	83.3%	257.0	51.4%

☐ SHOW SHIFTED COMPOSITE CURVES ΔTMIN 10

MINIMUM HEATING DEMAND

375.00 kW

MINIMUM COOLING DEMAND

500.00 kW

PINCH TEMPERATURE

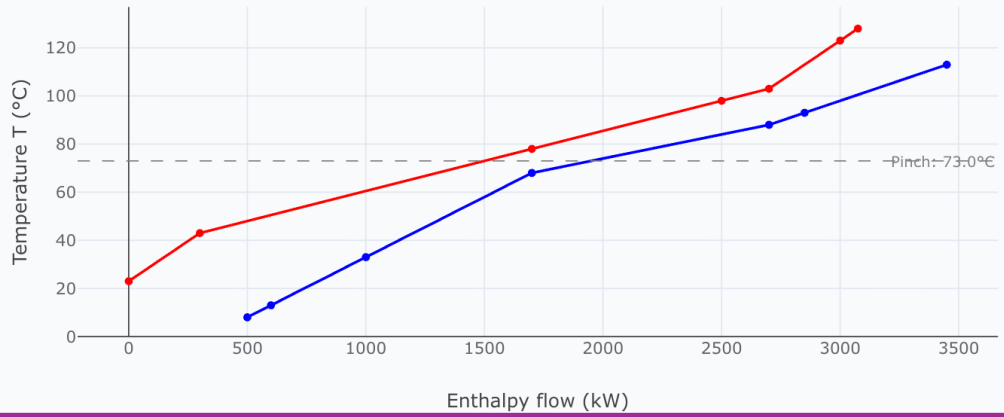
73.0 °C

HEAT RECOVERY POTENTIAL

2700.00 kW

Composite Curves

Hot streams (Heat sources) Cold streams (Heat sinks)



Grand Composite Curve



INTEGRATION WITH HEAT PUMP MODEL AND OPTIMIZATION

REFRIGERANT TYPE	MEDIUM SINK	COP	Q_SINK (KW)	Q_SOURCE (KW)	ACTUAL SINK TEMP [°C]	SHIFTED SINK (T+5K) [°C]	ACTUAL SOURCE TEMP [°C]	SHIFTED SOURCE (T-5K) [°C]	REFRIGERANT	HP STAGES
Natural	Water	4.34	375.0	288.6	113.0	118.0	58.6	53.6	Butane	2
Synthetic	Water	4.18	375.0	285.3	113.0	118.0	58.7	53.7	R1224yd	2

Click on any column header to sort. Click on any point in the chart to inspect its details and add it to the table.

MEDIUM SINK

All

REFRIGERANT TYPE

All

REFRIGERANT

All

HP STAGES

All

Select All

Clear Selection

☐ PLOT SOURCE-LIMITED POINTS (7)

THESE COVER ONLY PART OF THE DEMAND ABOVE T_SINK, SO THEY LIE OFF THE SINK PROFILE.

☐ PLOT DEMAND-LIMITED POINTS (9)

THESE COVER THE FULL DEMAND BUT LEAVE WASTE HEAT UNUSED AT T_SOURCE, SO THEY LIE OFF THE SOURCE PROFILE.

Feasible Combinations: 1635
shown in chart: 1619

COP SOURCE

f Use my own COP formula

Paste a custom COP to perform integration and optimization with your own heat pump.

